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Touchless glove application systems

Abstract

A glove application system includes a mounting bracket that provides an access aperture, and a glove cartridge assembly mountable to the mounting bracket and including a cartridge belt having a plurality of glove cartridges spaced along the cartridge belt, each glove cartridge having a glove releasably secured to the cartridge belt with a fastening system. The glove cartridge assembly is operable to advance the cartridge belt and sequentially align each glove cartridge with the access aperture. Extending a hand through the access aperture installs a glove from an aligned glove cartridge on the hand and simultaneously releases the glove from the cartridge belt.

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Primary Examiner: Hall; F Griffin*Attorney, Agent or Firm:* Vorys, Sater, Seymour and Pease LLP**Background/Summary****BACKGROUND**

(1) The use of disposable gloves (e.g., latex gloves, nitrile gloves, etc.) is commonplace in many industries, and is generally recognized as one of the primary safeguards against inadvertent or accidental cross-contamination between a user (wearer) and the surrounding environment. When used properly, gloves improve hygiene and can minimize exposure for both the user and others, especially those serviced by the person wearing the gloves.

(2) However, common errors that occur during glove application (placement) can compromise the overall effectiveness of glove use, such as allowing the exterior of the gloves to undergo significant physical contact prior to and during application. For example, gloves are commonly packaged in a manner that increases the probability that a user will physically contact multiple gloves with bare

hands upon extracting a single glove. Moreover, the first glove is typically applied with a non-gloved hand, thereby possibly contaminating the exterior of the first glove. In effect, simply putting disposable gloves on manually can inadvertently contaminate multiple gloves intended for sterile or sanitary applications. Moreover, gloves can be contaminated through inadvertent or intentional contact with non-sanitary surfaces prior to the intended use.

(3) Furthermore, in view of the COVID-19 pandemic that currently plagues our world, the use of disposable gloves has exponentially increased. Systems and methods of applying disposable gloves without increasing or propagating contamination is, therefore, greatly desired.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following figures are included to illustrate certain aspects of the present disclosure, and should not be viewed as exclusive embodiments. The subject matter disclosed is capable of considerable modifications, alterations, combinations, and equivalents in form and function, without departing from the scope of this disclosure.

(2) FIG. 1 is a front view an example glove application system, according to one or more embodiments.

(3) FIG. 2 is a front view another example glove application system, according to one or more additional embodiments.

(4) FIG. 3 is an exposed top view of the system of FIG. 1 or 2, according to one or more embodiments of the disclosure.

(5) FIG. 4 is a plan view of a portion of an example embodiment of the cartridge belt of FIG. 3, according to embodiments of the present disclosure.

(6) FIG. 5 is a top view of an example inflated glove, according to one or more embodiments.

(7) FIG. 6A is an isometric side view of another example glove application system, according to one or more embodiments.

(8) FIG. 6B is a schematic top view of the system of FIG. 6A.

(9) FIGS. 7A and 7B are side and front schematic views, respectively, of another example glove application system, according to one or more embodiments.

(10) FIG. 8 is a side schematic view of another example glove application system, according to one or more additional embodiments.

DETAILED DESCRIPTION

(11) The present disclosure relates to glove application systems for applying disposable gloves and, more particularly, to touchless glove application systems designed to minimize the potential for contamination of the exterior of the gloves during glove application (placement).

(12) The automated glove application systems described herein enable “touchless” self-donning (installation) of disposable gloves and thereby minimize or eliminate user contact with the external surfaces of the glove during the application process. The systems and methods discussed herein can enhance aseptic technique and increase the speed of glove application, while simultaneously preventing contamination associated with user error and inadvertent contaminating contact with the gloves. The glove application systems incorporate individually packaged and sterilized gloves that can be accessed by the user via an automated system. In some embodiments, individually packaged gloves may be provided in a pre-packaged, sterilized cartridge assembly that can easily be installed into the dispenser systems described herein. In one or more disclosed embodiments, the glove application systems described herein utilize a pressure system that provides cyclical negative and positive pressure conditions that aid in glove application and fitting the glove to the hand of a user.

(13) The presently-disclosed embodiments are advantageous in view of the current COVID-19 pandemic. In particular, personal protective equipment (PPE), such as disposable gloves, is prone

to rapid contamination and is typically discarded once contaminated, either on purpose or inadvertently. The automated glove application systems described herein can help reduce the amount of wasted disposable gloves that may be accidentally compromised, which would otherwise require the user to discard the contaminated glove and don or apply a new glove. The embodiments disclosed herein may be particularly advantageous in settings where a high degree of sanitation or sterilization is required or desired such as, but not limited to, hospitals, nursing homes, restaurants, bathrooms, etc. As used herein, the terms “don” and “donning” refer to the act of applying or installing a glove on a hand. The terms “don” and “donning” may be interchangeably used with terms such as “apply” or “install”.

(14) FIG. 1 is a front view of an example glove application system **100**, according to one or more embodiments. As illustrated, the glove application system **100** (hereafter “the system **100**”) may include a glove application module **102**. In some embodiments, the glove application module **102** may comprise a standalone system that may be portable to any desired location. In other embodiments, however, the glove application module **102** may be mounted to a rigid structure **104** for use. The rigid structure **104** may comprise a variety of stationary structures such as, but not limited to, a wall, a countertop, a stand, a public access point (e.g., a gas pump, a subway entrance, an elevator wall, etc.), or any combination thereof. Alternatively, the rigid structure **104** may comprise a movable (transportable) structure, such as a wheeled cart, an emergency vehicle, or any combination thereof.

(15) In some embodiments, a source of power may be included with the rigid structure **104** to operate the glove application module **102**, such as electrical grid power available through a wall power outlet (e.g., 110V), or electrical power derived from a vehicle power port, etc. In embodiments where the glove application module **102** is affixed to a piece of equipment, such as a gas pump, an elevator, etc., the electrical power may be derived from such equipment. Alternatively, or in addition thereto, the glove application module **102** may include a localized power source **106**, such as one or more batteries (e.g., 12V, 24V, etc.), fuel cells, a solar panel power system, or any combination thereof.

(16) The glove application module **102** includes a housing **108** that contains or supports various devices or mechanisms used to operate the glove application module **102**. In some embodiments, the housing **108** provides or otherwise defines a first or “install” cavity **110a** and a second or “pressure” cavity **110b** separated from the install cavity **110a** by a partition **112**. The cavities **110a,b** may alternately be referred to as “zones”. The partition **112** separates the first and second cavities **110a,b** but also defines a pressure vent **114** that provides fluid (air) communication between the first and second cavities **110a,b**.

(17) The glove application module **102** may also include a pressure system **116** operable to generate cyclical negative and positive pressure conditions in the install cavity **110a**. The negative and positive pressures are measured with respect to ambient or atmospheric pressure. Accordingly, negative pressure generation provided by the pressure system **116** may comprise any pressure condition within the install cavity **110a** that is lower than atmospheric pressure, and positive pressure generation may comprise any pressure condition within the install cavity **110a** that is higher than atmospheric pressure.

(18) In some embodiments, the pressure system **116** (shown in dashed lines) may be arranged within the pressure cavity **110b** and may be operable to generate negative and positive pressure conditions in the install cavity **110a** via the pressure vent **114**. In such embodiments, the pressure system **116** may generate cyclical negative and positive pressure within the pressure cavity **110b**, and such pressure changes will be communicated to the install cavity **110a** via the pressure vent **114**. More specifically, the pressure system **116** may be configured to generate a low pressure condition within the pressure cavity **110b** that draws air from the install cavity **110a** and into the pressure cavity **110b** via the pressure vent **114**. The pressure system **116** may then be configured to generate a high pressure condition within the pressure cavity **110b** that forces air into the install

cavity **110a** via the pressure vent **114**. In at least one embodiment, the install cavity **110a** may be substantially sealed to enhance the effects of the cyclical negative and positive pressure conditions. (19) Alternatively, the pressure system **116** may communicate directly with the pressure vent **114** via a hose or the like, thus not affecting pressure conditions in the pressure cavity **110b**, but only within the install cavity **110a**. In other embodiments, however, the pressure system **116** may be arranged outside of the pressure cavity **110b**, such as being attached to the exterior of the housing **104**. In such embodiments, the pressure cavity **110b** may be omitted as unnecessary, and the pressure system **116** may fluidly communicate directly with the install cavity **110a** through a sidewall of the housing **104**, for example, to generate the negative and positive pressure conditions in the install cavity **110a**.

(20) The pressure system **116** can include any suitable compressor, pump, or device capable of increasing and decreasing the pressure within the install cavity **110a**. Suitable examples for the pressure system **116** include, but are not limited to, a pressure piston, an air piston, a plunger, a wiper, a collapsible/deformable mechanism (e.g., a bellows, an accordion-type bellows, an expandable bladder, etc.), a reversing compressor or pump, or any combination thereof. In at least one embodiment, the pressure system **116** may comprise a spring-loaded bellows that may be actuated in one direction to create a positive pressure condition, and allowed to revert back to its natural state while creating a negative pressure condition. In some embodiments, air may be drawn in by the pressure system **116** from the exterior of the housing **108** to increase the pressure within the install cavity **110a**. In such embodiments, a vent **121** may be defined in a sidewall of the housing **108** and an inlet filter may be arranged within the vent **121** to filter the incoming air. The inlet filter may be a high efficiency particulate air (HEPA) filter configured to trap a variety of contaminants prior to entering the glove application module **102** and, more particularly, the install cavity **110a**.

(21) The glove application module **102** may further include a glove cartridge assembly **118** (shown generally in dashed lines) mounted to the housing **108** and including a plurality of individually packaged and releasably secured gloves. The gloves may comprise disposable gloves such as, but not limited to, latex gloves, nitrile gloves, butyl gloves, vinyl gloves, neoprene gloves, or any combination thereof. The gloves may be sterile or non-sterile. As discussed in more detail below, the gloves in the glove cartridge assembly **118** may be individually packaged and/or spaced along a cartridge belt configured to be driven (transferred) between opposing supply and recovery spools. The glove application module **102** may include means for selectively advancing individual gloves in the glove cartridge assembly **118** to align with an access aperture **120** defined in the housing **108**. The access aperture **120** provides a location where a user may be able to insert a hand into the install cavity **110a** via the access aperture **120** and simultaneously insert the hand into a glove aligned with the access aperture **120**.

(22) In some embodiments, as illustrated, the glove cartridge assembly **118** may be arranged within the install cavity **110a**. In other embodiments, however, the glove cartridge assembly **118** may be attached to the exterior of the housing **108** and the cartridge belt may be fed into the housing **108** on one side from an externally-mounted supply spool, and received by an externally-mounted recovery spool on an opposing side of the housing **108**.

(23) In some embodiments, the glove application module **102** may further include a user interface **122** designed to allow a user to operate the glove application module **102**. In some embodiments, the user interface **122** may comprise a touch interface, such as a self-sanitizing switch or button mounted to the housing **108**. In such embodiments, a user may manually flip the switch or press the button to activate operation of the glove application module **102**, which may cause the glove cartridge assembly **118** to advance the cartridge belt and align a glove with the access aperture **120**. In other embodiments, the user interface **122** may comprise a graphical user interface (GUI), such as a self-sanitizing touch-screen display that allows a user to operate the glove application module **102**. In such embodiments, the user interface **122** may be programmed to accept user input for

various glove selection criteria such as, but not limited to, glove size, glove color, glove material, glove style, right hand, left hand, or any combination thereof.

(24) In yet other embodiments, the user interface **122** may comprise a touchless interface. In such embodiments, for example, the touchless user interface **122** may comprise one or more sensors that facilitate touchless operation of the glove application module **102**. More particularly, various sensors incorporated into the glove application module **102** may sense movement or motion in proximity of the user interface **122** (e.g., waving a hand in front of the user interface **122**, approaching the glove application module **102**, etc.). Such movement or motion in proximity of the user interface **122** may cause the glove cartridge assembly **118** to advance and align a glove with the access aperture **120**. Suitable sensors for the user interface **122** include, but are not limited to, motion sensors (e.g., infrared (IR) sensors), photoelectric sensors (e.g., photo eyes), light curtains, light sensors, proximity sensors (e.g., radio-frequency identification (RFID) readers), or any combination thereof. In at least one embodiment, the touchless user interface **122** may comprise a proximity sensor employing RFID technology, for example, which may detect a user's identification (e.g., a company badge or the like) once in proximity of the glove application module **102**. In other embodiments, the touchless user interface **122** may comprise a voice-activated computer system with a microphone and speaker(s). In such embodiments, the user may speak voice commands to the user interface **122** to cause the glove cartridge assembly **118** to advance and align a glove with the access aperture **120**.

(25) In some embodiments, one or both of the pressure system **116** and the glove cartridge assembly **118** may be powered by the localized power source **106**. In other embodiments, one or both of the pressure system **116** and the glove cartridge assembly **118** may be powered with electrical grid power available through a wall power outlet (e.g., 110V), or electrical power derived from a vehicle power port. In yet other embodiments, however, a mechanical actuator **124** may form part of the user interface **122** and may be manually actuated by the user to operate the glove application module **102**. Such embodiments may prove advantageous in the event grid power or the localized power source **106** is unavailable, such as in remote or portable applications where ready access to electrical power may be limited.

(26) The mechanical actuator **124** may comprise, for example, one or more levers or pedals that a user must manually actuate (e.g., reciprocate) to operate the glove cartridge assembly **118** and/or the pressure system **116**. In the illustrated embodiment, the actuation system **124** comprises a lever that can be pivoted up or down by a user to operate the glove application module **102**. In at least one embodiment, the lever may comprise a “wrist blade” or the like, which comprises a generally flat surface that a user may actively engage with an arm, a wrist, etc. to manually actuate the glove cartridge assembly **118** and/or the pressure system **116** without having to use their hands. In other embodiments, the mechanical actuator **124** may comprise a treadle having a geared and/or pulley interconnection with one or both of the glove cartridge assembly **118** and the pressure system **116**, and user actuation (operation) of the treadle may cause the glove cartridge assembly **118** and/or the pressure system **116** to operate.

(27) Example operation of the glove application module **102** is now provided. In some embodiments, a user may activate operation of the glove application module **102** through interaction with the user interface **122** and/or the mechanical actuator **124**, as generally discussed above. In such embodiments, operating the user interface **122** (e.g., touch or touchless) and/or the mechanical actuator **124** may cause the glove cartridge assembly **118** to advance a cartridge belt and align a glove with the access aperture **120**. Once the glove is aligned with the access aperture **120** (or as the glove is being moved and aligned), the pressure system **116** may be activated to generate negative pressure (e.g., a vacuum) within the install cavity **110a**. The generated negative pressure may be sufficient to inflate the aligned glove into the interior of the install cavity **110a**, but each glove may be operatively coupled to the cartridge belt such that inflating the glove into the install cavity **110a** does not release the glove from the cartridge belt. In other embodiments,

however, the user must first penetrate a seal that covers the glove before the glove can be inflated into the install cavity **110a**, without departing from the scope of the disclosure.

(28) Once the glove is inflated and extended into the install cavity **110a**, the user may extend a hand into the inflated glove via the access aperture **120**, at which point the pressure system **116** may cycle (switch—return cycle) from generating negative pressure to generating positive pressure within the install cavity **110a**. Cycling the pressure system **116** from negative to positive pressure generation may be manually or electronically controlled. In embodiments where the pressure system includes a reciprocating device, such as a spring-loaded bellows, an air piston, or a treadle, cycling the pressure system **116** from negative to positive pressure generation may naturally occur during operation. In some embodiments, the duration of the negative/positive pressure cycle may be based on a selected timing, such as every 2 seconds, 3, seconds, 5 seconds, etc. In other embodiments, cycling the pressure system **116** from negative to positive pressure may be triggered by operation of one or more internal sensors that detect the presence of the hand. In such embodiments, a motion sensor or a light curtain may cause the pressure system **116** to switch (cycle) from negative to positive pressure within the install cavity **110a** once the presence of a hand within the inflated glove is detected. In some embodiments, the pressure system **116** may continuously operate, either cycling between negative and positive pressure conditions, or maintaining a negative pressure within the install cavity **110a**. In other embodiments, however, the pressure system **116** may only be operated as needed to install a glove, following which operation of the pressure system **116** may cease.

(29) The positive pressure generated by the pressure system **116** may help seat the glove onto the user's hand and thereby provide tactile feedback to the user that the glove is properly donned (installed). Without the positive pressure forces that help seat the glove on the user's hand, a user may be able to withdraw the hand without removing the glove, thus defeating the purpose of the glove application module **102**. Accordingly, once the user feels that the glove is properly applied (seated) on the hand from the positive pressure condition, the now-gloved hand may be withdrawn from the access aperture **120**. In some embodiments, withdrawing the gloved hand from the access aperture **120** may release the glove from the cartridge belt. In other embodiments, or in addition thereto, the positive pressure condition may create sufficient detachment forces to release the glove from the cartridge belt.

(30) In some embodiments, withdrawing the gloved hand from the install cavity **110a** may trigger operation of the glove cartridge assembly **118** to advance and align another glove with the access aperture **120**. In such embodiments, the default position for the glove application module **102** may be to have a glove already advanced and aligned with the access aperture **120** before a user approaches the system **100**. Accordingly, once a glove is dispensed from the glove application module **102** and the user withdraws the hand, the glove cartridge assembly **118** may automatically advance and align another glove with the access aperture **120**, thus readying the glove application module **102** for subsequent and continued use.

(31) FIG. 2 is a front view of another example glove application system **200**, according to one or more additional embodiments. The glove application system **200** (hereafter “the system **200**”) may be similar in some respects to the system **100** of FIG. 1 and therefore will be best understood with reference thereto, where like numerals will refer to like components not described again. As illustrated, the system **200** includes a plurality of vertically stacked glove application modules, shown as glove application modules **102a**, **102b**, and **102c**. While three glove application modules **102a-c** are depicted in FIG. 2, more or less than three may be employed in the system **200**, without departing from the scope of the disclosure. The glove application modules **102a-c** may be mounted to the rigid structure **104** for use, or may alternatively comprise a standalone unit.

(32) In some embodiments, each glove application module **102a-c** may include a separate housing **108** that contains or supports various devices or mechanisms used to operate the glove application modules **102a-c**. In other embodiments, however, the glove application modules **102a-c** may share

a common housing that incorporates or includes all of the glove application modules **102a-c**. Each glove application module **102a-c** may provide an install cavity **110a** and an access aperture **120** providing a location where the user is able to insert a hand into the corresponding install cavity **110a** and simultaneously into an inflated glove aligned with the corresponding access aperture **120**. (33) In some embodiments, the system **200** may further include a mechanical housing **202**, which may form part of the housing(s) **108** or may be coupled thereto. The mechanical housing **202** may house the pressure system **116** and various electrical components (e.g., computer systems, etc.) configured to operate the glove application modules **102a-c**. In other embodiments, however, the pressure system **116** may alternatively be coupled to the exterior of the housing(s) **108**. In such embodiments, the mechanical housing **202** may be replaced with an additional glove application module. The pressure system **116** may be in fluid communication with the install cavity **110a** of each glove application module **102a-c**. In operation, and depending on user input, the pressure system **116** may be operable to selectively generate cyclical negative and positive pressure in any or all of the install cavities **110a**.

(34) Each glove application module **102a-c** may further include separate and discrete glove cartridge assemblies **118** (shown in dashed lines) mounted to the housing(s) **108**. Each glove cartridge assembly **118** may be pre-packaged and sterilized prior to installation, and may include a plurality of gloves individually packaged and releasably secured to a cartridge belt configured to be driven (transferred) between opposing supply and recovery spools. Each glove cartridge assembly **118** may also be programmed or otherwise operable to selectively advance individual gloves to the corresponding access aperture **120**, and depending on user input. More specifically, the system **200** may further include one or more user interfaces **122** (three shown) designed to allow a user to operate the glove application modules **102a-c**. In the illustrated embodiment, each glove application module **102a-c** includes a separate user interface **122**. In such embodiments, each user interface **122** may be configured to trigger operation of the corresponding glove application module **102a-c**. In other embodiments, however, the system **200** may include a single user interface **122** configured to selectively operate any of the glove application modules **102a-c**, depending on user input.

(35) In some embodiments, each glove application module **102a-c** may be configured to dispense a dissimilar type of glove. For example, each glove cartridge assembly **118** may include a different type of glove, and a user may be able to select a desired type of glove by activating operation of the particular glove application module **102a-c** via the corresponding user interface **122**. The criteria for the different types of gloves that may be dispensed in the system **200** include, but are not limited to, size, color, material, style, right hand, left hand, or any combination thereof.

(36) In some embodiments, the system **200** may further include a mechanical actuator **204** that may be manually actuated by the user to operate one or both of the pressure system **116** and the glove cartridge assemblies **118**. In the illustrated embodiment, the mechanical actuator **204** comprises a spring-loaded, foot pedal bellows configured to help operate or replace the pressure system **116**. More specifically, the bellows may be pressed down by the user (e.g., with a foot) and subsequently allowed to expand to create cyclical positive and negative pressures within the install cavities **110a**. Such embodiments may prove advantageous for remote or portable applications, where ready access to power for the pressure system **116** may be limited.

(37) Operation of the glove application modules **102a-c** may be generally similar to the operation of the glove application module **102** of FIG. 1 above, except for the ability to dispense different types of gloves via a plurality of access apertures **120**. Moreover, however, in some embodiments, the system **200** may further include an optional sterilization module **206** coupled to or forming part of the housing(s) **108**. As illustrated, the sterilization module **206** may provide or define an aperture **208** into which a user can place a gloved hand (or hands) to sanitize and/or sterilize the glove(s). The sterilization module **206** can include a sterilizing device **210** operable to sanitize and/or sterilize the glove(s) upon introduction into the aperture **208**. The sterilizing device **210** may

include, but is not limited to, an ultraviolet (UV) lamp (e.g., a short-wavelength UV lamp, a UV-C lamp, a germicidal UV lamp, etc.), an ozonator, a chemical disinfectant (e.g., a sprayer, an aerosolizer, a bath, etc.), or any combination thereof. The sterilization module **206** may prove advantageous sterilizing gloves that have been in limited (or longer) use, thus minimizing the need for repeated glove removal and application of new gloves, and thereby helping to reduce wasted PPE.

(38) In some embodiments, the sterilization module **206** may be selectively actuatable via one of the user interfaces **122**, thus allowing a user to selectively activate the sterilizing device **210** on demand. In other embodiments, however, the sterilization module **206** may include a sensor **212** (e.g., a proximity sensor) configured to detect the presence of a hand near or within the aperture **208**. Once a hand is detected, operation of the sterilizing device **210** may be triggered. In yet other embodiments, operation of the sterilizing device **210** may be continuous and the user need only insert the gloved hand into the aperture **208**, without departing from the scope of the disclosure.

(39) FIG. **3** is an exposed top view of the system **100** of FIG. **1**, according to one or more embodiments of the disclosure. As illustrated, the glove cartridge assembly **118** may include a supply spool **302a**, a recovery spool **302b**, and a radial cartridge belt **304** extendable between the supply and recovery spools **302a,b**. The cartridge belt **304** may be preloaded with a plurality of individually spaced and separated glove cartridges **306** attached to the cartridge belt **304** at predetermined locations and spacing. Each glove cartridge **306** includes a single glove and a means of attaching the glove to the cartridge belt **304**. In some embodiments, the glove may be vacuum-packed and sealed within sterilized packaging, but could alternatively not include any packaging and instead merely be attached to the cartridge belt **304**. The cartridge belt **304** can have a stipulated number of glove cartridges **306** affixed thereto such as, but not limited to, 25, 50, 100, 200, any number between 25 and 200, less than 25, or more than 200.

(40) Before the glove cartridge assembly **118** is installed in or on the glove application module **102**, the cartridge belt **304** may be sterilized and pre-packaged for delivery. In some embodiments, the pre-packaged and sterilized cartridge belt **304** may be entirely (or mostly) pre-wound onto the supply spool **302a**, but could alternatively be installed on an existing supply spool **302a** forming part of the system **100**.

(41) In some embodiments, installing the glove cartridge assembly **118** may entail merely attaching the glove cartridge assembly **118** to the glove application module **102** such that the cartridge belt **304** extends across the access aperture **102**. In such embodiments, the glove cartridge assembly **118** may be provided to the user with the cartridge belt **304** already extending between the supply and recovery spools **302a,b**. In other embodiments, however, it is contemplated herein that a user will be able to install the glove cartridge assembly **118**. In such embodiments, one end of the cartridge belt **304** is pulled radially from the supply spool **302a**, extended (threaded) through an alignment guide **308** provided within the housing **108**, and operatively coupled to the recovery spool **302b** such that driving rotation of the recovery spool **302b** will correspondingly pull additional length of the cartridge belt **304** from the supply spool **302a** and through the alignment guide **308**. As illustrated, the alignment guide **308** is aligned with the access aperture **120**, and advancing (pulling) the cartridge belt **304** through the alignment guide **308** will allow individual glove cartridges **306** to be sequentially aligned with the access aperture **120** for operation.

(42) In some embodiments, the recovery spool **302b** may be driven in rotation by a motor **310** configured to rotate the recovery spool **302b** about a rotational axis **312**. In such embodiments, the motor **310** may be operated to advance the cartridge belt **304** through the alignment guide **308** at predetermined increments that accurately and precisely align individual glove cartridges **306** with the access aperture **120**. In other embodiments, however, or in addition thereto, the glove application module **102** may further include one or more pinion gears **314** engageable with the cartridge belt **304** and rotatable to advance the cartridge belt **304** through the alignment guide **308** at predetermined increments to align individual glove cartridges **306** with the access aperture **120**.

In such embodiments, gear teeth of the pinion gear **314** may mate with corresponding holes defined in the cartridge belt **304** such that rotation of the pinion gear **314** drives the gear teeth against the corresponding holes and advances the cartridge belt **314**. Moreover, in such embodiments, the recovery spool **302b** may not be required, but instead the used portion of the cartridge belt **304** may self-coil on the opposite side of the housing **108** past the pinion gear(s) **314**.

(43) Those skilled in the art, however, will readily appreciate that various other ways may be employed to advance the cartridge belt **304**, without departing from the scope of the disclosure. For example, as indicated above, in at least one embodiment, a mechanical actuator **124** (FIG. 1) may be included and manually actuated by the user to operate the glove application module **102**; e.g., the glove cartridge assembly **118**.

(44) In example operation, the cartridge belt **304** is advanced from the supply spool **302a** until an individual glove cartridge **306** aligns with the access aperture **120**. In some embodiments, the supply spool **302a** may be spring loaded and thereby generate constant tension in the cartridge belt **304**. In at least one embodiment, for example, the supply spool **302a** may include a torsion spring **316** that maintains a tensile load (pull) on the cartridge belt **304** and thereby keeps the cartridge belt **304** taut at all times.

(45) Once a glove cartridge **306** is properly aligned with the access aperture **120**, or as the glove cartridge **306** is moving to the access aperture **120**, the pressure system **116** (FIGS. 1 and 2) may be operated (activated) to generate negative pressure within the install cavity **110a**. The negative pressure may be sufficient to at least partially inflate a glove **318** from the aligned glove cartridge **306** such that the glove **318** extends into the install cavity **110a**. The negative pressure required to inflate the glove **318**, however, may not be sufficient to release the glove **318** from the cartridge belt **304**. In some embodiments, aligning the glove cartridge **306** with the access aperture **120** forms a barrier between the access aperture **120** and the install cavity **110a**, and thereby forms a pressure-sensitive seal that helps inflate the glove **318**. The glove **318** may be inflated from a packaged (stowed) configuration, which may include a packaged and/or compressed glove with or without an exterior seal or covering.

(46) Once the glove **318** is inflated, the user may extend a hand into the inflated glove via the access aperture **120**, and the pressure system **116** may then cycle (switch) to generating positive pressure in the install cavity **110a**. As mentioned above, the positive pressure can force the glove **318** against the user's hand and thereby provide positive tactile feedback that lets the user know that the glove **318** is properly donned (applied) and ready to be withdrawn. Once the user feels that the glove **318** is properly applied, the now-gloved hand may be withdrawn from the access aperture **120**, thus releasing the glove **318** from the cartridge belt **304**. In some embodiments, however, and as also mentioned above, the positive pressure within the install cavity **110a** may be sufficient to release of the glove **318** from the cartridge belt **304**.

(47) Upon withdrawing the now-gloved hand from the install cavity **110a**, the cartridge belt **304** may be advanced until another (subsequent) glove cartridge **306** aligns with the access aperture **120**. Alternatively, the cartridge belt **304** may be advanced upon re-activating the glove application module **102**. In either scenario, advancing the cartridge belt **304** could be done either manually or through an automated system, as generally described herein. The recovery spool **302b** may be configured to collect and/or wind the used cartridge belt **304** and corresponding empty apertures **320** where individual glove cartridges **306** were previously provided. Once the supply spool **302a** is depleted of glove cartridges **306**, the glove cartridge assembly **118** may be replaced with a new glove cartridge assembly. The glove application module **102** may be configured with interchangeable or refillable glove cartridge assemblies that enable quick refills that minimize potential contamination of the packaged gloves **318**. The glove cartridge assembly **118** and any replacement glove cartridge assemblies may come pre-packaged, sterilized, and sealed prior to installation.

(48) In some embodiments, the system **100** may further include a sterilizing device **322** positioned

within the install cavity **110a**. The sterilizing device **322** may be similar in some respects to the sterilizing device **210** of FIG. 2 and, therefore, may comprise any of the examples given above for the sterilizing device **210**. The sterilizing device **322** may be operable to sanitize and/or sterilize the glove **318** upon introduction (inflation) into the install cavity **110a**. In some embodiments, operation of the sterilizing device **322** may be activated once the inflated glove **318** is detected within the install cavity **110a** with one or more sensors **324** similar to the sensor(s) **212** of FIG. 2.

(49) FIG. 4 is a plan view of a portion of an example embodiment of the cartridge belt **304**. As illustrated, the cartridge belt **304** may comprise a generally planar strip of material having a plurality of glove cartridges **306** coupled thereto. The cartridge belt **304** may be made of a variety of flexible materials, such as a mechanically durable plastic (e.g., vinyl, polyethylene, etc.), but could alternatively be made of paper or a paper product. In at least one embodiment, however, the cartridge belt **304** may be made of the same material as the gloves **318**.

(50) Each glove cartridge **306** may include a glove **318** individually packaged (stowed) and releasably secured to the cartridge belt **304** in the packaged configuration. In some embodiments, each glove **318** may be vacuum packed within a sealed interface **402**, such as a protective cellophane seal. In at least one embodiment, the sealed interface **402** may provide tamper proof packaging, but may be thin enough that it will easily tear upon user penetration with a hand. In other embodiments, however, the sealed interface **402** may be omitted. In either scenario, each glove **318** may be sterilized prior to installation on the cartridge belt **304**, thus providing a plurality of sealed and sterilized replacement glove cartridges **306** spaced from each other along the cartridge belt **304** at known distances.

(51) Each glove cartridge **306** may further include a fastening system **404** configured to releasably secure the glove **318** to the cartridge belt **304** in the corresponding glove cartridge **306**. In some embodiments, for example, the fastening system **404** may be configured to receive and releasably secure the wrist portion of each glove **318** (e.g., the wrist bead), which helps restrain the glove **318** during inflation. In at least one embodiment, the fastening system **404** may comprise a split O-ring that releasably secures the wrist bead of the glove **318**. In other embodiments, however, the fastening system **404** may comprise a releasable adhesive, a wax, a Velcro attachment, one or more mechanical staples, or any other type of releasable mechanism that can hold the glove **318** in place until the proper amount of tension is applied on the glove **318** to release it from the fastening system **404**. In yet another embodiment, the fastening system **404** may comprise overlaying substrates and the wrist bead of the glove **318** may interpose the overlaying substrates in a sandwich-type arrangement that releasably secures the glove **318**. The fastening system **404** may also facilitate a pressurizable seal that allows the glove **318** to be inflated into the install cavity **110a** (FIGS. 1-3).

(52) In embodiments where the glove application module **102** (FIG. 3) includes the pinion gear(s) **314** (FIG. 3), the cartridge belt **304** may define a series of equidistantly spaced holes **406** along one or both of its lateral sides. The holes **406** may be configured to mate with the gear teeth of the pinion gear **314** and, as the pinion gear **314** rotates, the gear teeth sequentially engage the holes **406** and thereby advance (drive) the cartridge belt **304** forward. Accordingly, the holes **406** may comprise equidistantly spaced notches or other indexing features configured to aid in the timing and accurate placement of the glove cartridges **306** during operation.

(53) In some embodiments, the cartridge belt **304** may provide or otherwise define spacer sections **408** interposing longitudinally adjacent glove cartridges **306** along the length of the cartridge belt **304**. During example operation in accordance with some embodiments, the cartridge belt **304** may be advanced until a spacer section **408** is aligned with the access aperture **120** (FIGS. 1-3). The aligned spacer section **408** may be configured to limit access to the install cavity **110a** (FIGS. 1-3) between glove application cycles to minimize contamination. In at least one embodiment, the spacer section **408** may form a seal at the access aperture **120**, which may help prevent entry of particulates and contaminants between glove application cycles.

(54) FIG. 5 is a top view of an example inflated glove **318**, according to one or more embodiments. As illustrated, the glove **318** may be releasably coupled to the cartridge belt **304** with the fastening system **404**, as generally described above. In the illustrated example, the fastening system **404** may be fixed to the cartridge belt **304** and may be configured to receive and secure a rolled wrist bead **502** of the glove **318**.

(55) As illustrated, the fastening system **404** can comprise a split O-ring configured to receive the wrist bead **502** and thereby releasably secure the glove **318** to the cartridge belt **304**. More specifically, the split O-ring may comprise a tubular structure **504** that defines an interior **506**. In some embodiments, for example, the split O-ring may comprise a length of tubing or hose, and may be made of a variety of materials including, but not limited to, an elastomer, a polymer, or a composite material. The tubular structure **504** may or may not form a complete circle. A split or opening **508** may be defined about the entire tubular structure **504** to enable the wrist bead **502** to be received within the interior **506** to releasably secure the glove **318** to the cartridge belt **304**. The tubular structure **504** may be configured to release the wrist bead **502** when the glove **318** receives the hand of a user, when the user withdraws the glove from the install cavity **110a** (FIGS. 1-3), or upon experiencing positive pressure within the install cavity **110a**.

(56) As will be appreciated, other types of quick-release fastening systems **404** may also be employed including, but not limited to, a frangible seal, a frangible adhesive, a releasable interference fit, a releasable mechanical connection, or any combination thereof. As mentioned above, the attachment force provided by the fastening system **404** may be sufficient such that the negative pressure within the install cavity **110a** that inflates the glove **318** does not cause the glove **318** to release from the fastening system **404**.

(57) FIG. 6A is an isometric side view of another example glove application system **600**, according to one or more embodiments. The glove application system **600** (hereafter “the system **600**”) may be similar in some respects to the system **100** of FIG. 1. As illustrated, the system **600** includes a glove application module **602** that may comprise a standalone, portable system that can be situated on a rigid surface, such as a countertop the like. The glove application module **602** includes a housing **604** that provides or otherwise defines an install cavity **606**. In some embodiments, the housing **604** may comprise a cylindrical tube having a circular cross-section, and the install cavity **606** is defined within the tube. Moreover, all or a portion of the housing **604** may be made of a transparent material, such as glass or clear plastic, to enable a user to view the install cavity **606** during operation. An access aperture **608** may be defined or otherwise provided at one end of the housing **604** and provides a location where a user may insert a hand into the install cavity **606** and simultaneously into a glove (not shown) aligned with the access aperture **608**.

(58) The glove application module **602** may further include a pressure system **610** operable to generate cyclical negative and positive pressure conditions in the install cavity **606**. In at least one embodiment, the pressure system **610** may comprise a collapsible/deformable mechanism, such as a bellows (e.g., accordion-type, spring-loaded, etc.). In example operation, the bellows may be actuated in one direction to create a negative pressure condition within the install cavity **606** and actuated in the opposite direction to create a positive pressure condition within the install cavity **606**.

(59) FIG. 6B is a schematic top view of the system **600**, according to one or more embodiments. As illustrated, the glove application module **602** may further include a glove cartridge assembly **612** that may be mounted to the housing **604**. In some embodiments, the glove cartridge assembly **612** may include a supply spool **614a**, a recovery spool **614b**, and a radial cartridge belt **616** extendable between the supply and recovery spools **614a,b**. Similar to the cartridge belt **304** of FIG. 3, the cartridge belt **616** may be preloaded with a plurality of individually spaced and separated glove cartridges **618** attached to the cartridge belt **616** at predetermined locations and spacing. Each glove cartridge **618** includes a single glove and a means of attaching the glove to the cartridge belt **616**.

(60) In some embodiments, the glove cartridge assembly **612** may be installed by attaching the

glove cartridge assembly **612** to the glove application module **602** such that the cartridge belt **616** extends across the access aperture **608**. In such embodiments, the glove cartridge assembly **612** may be provided to the user with the cartridge belt **616** already extending between the supply and recovery spools **614a,b**. In other embodiments, however, it is contemplated herein that a user will install the glove cartridge assembly **612** by pulling one end of the cartridge belt **616** from the supply spool **614a**, across (in front of) the access aperture **608**, and operatively coupling the end of the cartridge belt **616** to the recovery spool **614b** such that driving rotation of the recovery spool **614b** will correspondingly pull additional length of the cartridge belt **616** from the supply spool **614a**.

(61) Advancing the cartridge belt **616** from the supply spool **614a** to the recovery spool **614b** will allow individual glove cartridges **618** to be sequentially aligned with the access aperture **608**. In some embodiments, the recovery spool **614b** may be similar to the recovery spool **302b** of FIG. 3 and, therefore, may be driven in rotation by a motor (not shown). In other embodiments, however, one or more pinion gears (not shown) may advance the cartridge belt **616** to align individual glove cartridges **618** with the access aperture **608**. In yet other embodiments, the system **600** may include a mechanical actuator **620**, such as a wrist blade or ratchet lever that may be manually actuated by the user to operate the glove application module **602** and otherwise advance the cartridge belt **616**. In some embodiments, actuating the mechanical actuator **620** may also initiate operation of the pressure system **610**.

(62) In example operation, the cartridge belt **616** is advanced from the supply spool **614a** until an individual glove cartridge **618** aligns with the access aperture **608**. Once a glove cartridge **618** is properly aligned with the access aperture **608**, or as the glove cartridge **618** is moving to the access aperture **608**, the pressure system **610** may be operated (activated) to generate negative pressure within the install cavity **606**. Similar to the pressure system **116** of FIGS. 1 and 2, the pressure system **610** may be automated or otherwise powered using a mechanical actuator (e.g., a spring-loaded, foot pedal bellows) that is manually actuated by the user.

(63) In at least one embodiment, the pressure system **610** may communicate with the install cavity **606** via a pressure vent **622**. The negative pressure generated by the pressure system **610** may be sufficient to inflate a glove **624** from an aligned glove cartridge **618** such that the glove **624** extends into the install cavity **606**. Once the glove **624** is inflated, the user may extend a hand into the inflated glove via the access aperture **608**. In embodiments where the housing **604** is made of a transparent material, the user may be able to view this process and see where the hand is being inserted. The pressure system **610** may then cycle (switch) to generating positive pressure in the install cavity **606**, which may force the glove **624** against the user's hand and thereby provide positive tactile feedback. The positive pressure may also be sufficient to release of the glove **624** from the cartridge belt **616**. The user may then withdraw the now-gloved hand.

(64) FIGS. 7A and 7B are side and front schematic views, respectively, of yet another example glove application system **700**, according to one or more embodiments. Similar to the other glove application systems described herein, the glove application system **700** (hereafter “the system **700**”) may be configured to sequentially and systematically dispense and don (apply) disposable gloves. Unlike the prior disclosed systems, however, the system **700** does not require the use of a pressure system, or cyclical positive and negative pressure conditions.

(65) In some embodiments, as illustrated, the system **700** may be mounted to a rigid structure **702**, such as a gas pump, a subway entrance, a restroom wall, or any other location where rapid and discrete disposable glove dispensing may be desired. In some embodiments, the system **700** may include a mounting bracket **704** that may secure the system **700** to the rigid structure **702** in a desired orientation. The mounting bracket **704** may be fixed to the rigid structure **702** using one or more mechanical fasteners or an adhesive, for example.

(66) As illustrated, the system **700** may include a glove cartridge assembly **706** that may be mounted to the mounting bracket **704**. In some embodiments, the glove cartridge assembly **706**

may form an integral part of the system **700**. In other embodiments, however, the glove cartridge assembly **706** may alternatively comprise a pre-packaged and sterilized container including a supply of disposable gloves that can be mounted to the system **700** (e.g., the mounting bracket **704**). In such embodiments, when the supply of gloves is depleted from the glove cartridge assembly **706**, the glove cartridge assembly **706** may be detached from the system **700** and replaced with a new glove cartridge assembly.

(67) In some embodiments, the glove cartridge assembly **706** may include a supply spool **708** and a radial cartridge belt **710** (shown in dashed lines in FIG. 7B) wound onto the supply spool **708**. The cartridge belt **710** may be preloaded with a plurality of individually spaced and separated glove cartridges **712** (FIG. 7A) attached to the cartridge belt **710** at predetermined locations and spacing. Each glove cartridge **712** includes a single glove **714** (FIG. 7B) and a means of attaching the glove **714** to the cartridge belt **710**; e.g., the fastening system **404** of FIGS. 4 and 5. In at least one embodiment, for example, each glove **714** may be attached to the cartridge belt **710** using the split O-ring embodiment described with reference to FIG. 5.

(68) To install the glove cartridge assembly **706**, one end of the cartridge belt **710** may be pulled radially from the supply spool **708** and across (in front of) an access aperture **716** provided by the mounting bracket **704**. In at least one embodiment, the cartridge belt **710** may be extended (threaded) through an alignment guide **718** attached to the mounting bracket **704** and configured to align successive glove cartridges **712** with the access aperture **716**. As illustrated, the alignment guide **718** is aligned with the access aperture **716**, and advancing (pulling) the cartridge belt **710** through the alignment guide **718** will allow individual glove cartridges **712** to be sequentially aligned with the access aperture **716** for operation.

(69) In some embodiments, the system **700** may include a recovery spool (not shown) where the used cartridge belt **710** may be wound after dispensing its gloves **714**. In other embodiments, however, the recovery spool may be omitted and the used cartridge belt **710** may alternatively collect (e.g., self-coil) at the bottom of the system **700**. In such embodiments, portions of the used cartridge belt **710** may be severed or removed from time to time. The cartridge belt **710** may include, for example, pre-perforated tear lines defined in the cartridge belt **710** at predetermined locations, and the cartridge belt **710** may be severed at the tear lines.

(70) In some embodiments, the system **700** may be motorized and otherwise powered to advance the cartridge belt **710** through the alignment guide **718** at predetermined increments that accurately and precisely align individual glove cartridges **712** with the access aperture **716**. In such embodiments, the power required to operate the system **700** may be derived from the rigid structure **702**. In other embodiments, however, the system **700** may be manually actuated by the user. In such embodiments, the system **700** may include a mechanical actuator **720** that may be manually manipulated by a user to advance the cartridge belt **710** until a glove cartridge **712** aligns with the access aperture **716**. In the illustrated embodiment, the mechanical actuator **720** comprises a wrist blade or ratchet lever that a user may engage with a hand, wrist, or forearm to manually advance the cartridge belt **710**. In such embodiments, the gearing and/or mechanics of the mechanical actuator **720** may be designed such that with each actuation cycle of the mechanical actuator **720** a new glove cartridge **712** is advanced and aligned with the access aperture **716**.

(71) In example operation, the cartridge belt **710** is advanced from the supply spool **708** until an individual glove cartridge **712** aligns with the access aperture **716** at the alignment guide **718**. Once the glove cartridge **712** is properly aligned with the access aperture **716**, the user may extend a hand **722** (FIG. 7A) into the glove cartridge **712** via the access aperture **716** and simultaneously into the glove **714** packaged in the glove cartridge **712**. The act of extending the hand **722** into the glove cartridge **712** causes the glove **714** to expand from its stowed configuration and be received onto the hand **722**. Moreover, in some embodiments, extending the hand **722** into the glove **714** may release the glove **714** from the cartridge belt **710**. More specifically, extending the hand **722** into the glove **714** provides over pressure that releases the glove **714** from the means of attaching

the glove **714** to the cartridge belt **710**; e.g., the fastening system **404** of FIGS. **4** and **5**. The now-gloved hand **722** may then be withdrawn from the access aperture **716**.

(72) In some embodiments, the system **700** may further include a user interface **724** (FIG. **7B**) designed to allow a user to operate the system **700**. In embodiments where the system **700** is attached to a gas pump, the user interface **724** may be the same interface used to operate the gas pump. In such embodiments, the user interface **724** may be in communication with the system **700** and may comprise any of the touch or touchless configurations described for the user interface **122** of FIGS. **1-2**.

(73) In some embodiments, the system **700** may be operable to dispense a glove for a fee. In such embodiments, the user interface **724** may be used to handle user payments and trigger operation of the system **700** once the proper fee has been rendered. In other embodiments, however, the user interface **724** may form an integral part of the system **700**, without departing from the scope of the disclosure.

(74) FIG. **8** is a side schematic view of yet another example glove application system **800**, according to one or more additional embodiments. The glove application system **800** (hereafter “the system **800**”) may be configured to provide a user with a means of donning (installing) a single glove without contaminating the glove. As illustrated, the system **800** includes an applicator **802** that defines an aperture **804**, and a glove **806** is releasably mounted to the applicator **802** at the aperture **804** with a fastening system **808**. The applicator **802** may comprise a generally planar structure made of a rigid or semi-rigid material such as, but not limited to, cardboard, cardstock, a plastic, a composite material, a metal, or any combination thereof. In some embodiments, the applicator **802** may be disposable after use, but could alternatively be reusable or recyclable.

(75) While the applicator **802** is depicted in FIG. **8** as exhibiting a particular shape or design, the shape of the applicator **802** may vary, without departing from the scope of the disclosure, and as long as the applicator **802** is large enough to accommodate the glove **806** and the fastening system **808**. In some embodiments, for example, the applicator **802** may exhibit a substantially rectangular or square shape, or any other polygonal shape. In other embodiments, the applicator **802** may exhibit a generally circular, oval, or ovoid shape.

(76) As shown in FIG. **8**, the system **800** may be provided to the user with the glove **806** in a packaged (stowed) configuration. In the stowed configuration, the glove **806** may be compressed or flattened to about the thickness of the applicator **802**, and may or may not include an exterior seal or covering.

(77) The fastening system **808** may be similar to any of the fastening systems described herein and configured to releasably secure the glove **806** to the applicator **802**. In at least one embodiment, for instance, the fastening system **808** may comprise a type of split O-ring fastening system similar to the embodiment described with reference to FIG. **5**. In other embodiments, however, the fastening system **808** may comprise a releasable or frangible adhesive or seal, without departing from the scope of the disclosure. The fastening system **808** may secure the glove **806** to the applicator **802** until a user inserts a hand into the glove **806** with enough force to release the glove **806** from the applicator **802**.

(78) In some embodiments, the applicator **802** may provide or define a tab **810** that provides a location where a user may grasp the applicator **802** in preparation for donning (installing) the glove **806**. In other embodiments, however, the tab **810** may be omitted and the user may simply grasp a portion of the applicator **802** to don the glove **806**.

(79) In example operation, the user grasps the applicator **802** at the tab **810** (or another portion of the applicator **802**) with one hand **812** and simultaneously inserts the other hand (not shown) into the glove **806** at the aperture **804**. The user advances the other hand through the aperture **804** and into the glove **806** until sufficient tension is applied on the glove **806** to release it from the fastening system **808**. Once the glove **806** is released from the applicator **802**, the user may remove the now-gloved hand and dispose of the applicator **802** as appropriate (e.g., trash or recycle). In

some applications, the user can repeat the donning process for the first hand **812** using a new system **800** and a new glove **806** and thereby install the new glove **806** on the first hand **812**.

(80) In some embodiments, the glove **806** may remain attached to the applicator **802** for use. More specifically, in some cases, it may be desired to have the applicator **802** remain at or near a user's wrist to operate as a type of guard or shield. For example, the applicator **802** may shield the user from splashes or droplets of fuel while using a gas pump. In such embodiments, the user may advance the hand through the aperture **804** and into the glove **806**, but not far enough to release the glove **806** from the applicator **802**. When it is desired to remove the glove **806**, the user can grasp the applicator **802** again (e.g., at the tab **810** or another portion of the applicator **802**) and draw the hand out of the glove **806**.

(81) It is contemplated herein that the system **800** may be available to users in a package or device that contains a plurality of systems **800**. In such embodiments, the tab **810** of each system **800** may be easily accessible by the user to remove a given system **800** from the plurality. In at least one embodiment, for example, a plurality of the systems **800** may be arranged in a stacked relationship and dispensed from a manual or automated dispensing machine (not shown) that sequentially dispenses a new system **800** upon removal of the prior system **800**. In other embodiments, the systems **800** may be stacked within a static dispenser (not shown), such as a box, and the user may be able to access the systems **800** within the static dispenser as needed.

(82) It is also contemplated herein that the system **800** may form an integral piece or part of any of the systems **100**, **200**, **300**, **600**, or **700** described herein. More specifically, in some embodiments, a plurality of the systems **800** may be coupled to a cartridge belt and spaced from each other to form corresponding glove cartridges, or the plurality of systems **800** may alternatively be attached side-by-side to form the cartridge belt itself. The cartridge belt may then be sequentially advanced to align a given glove **806** with an install cavity and operation of a pressure system may be triggered to inflate the glove **806** into the install cavity.

(83) Embodiments disclosed herein include:

(84) A. A glove application system that includes a glove application module providing a housing that defines an install cavity and an access aperture providing access into the install cavity, a pressure system operable to generate negative pressure and positive pressure in the install cavity, and a glove cartridge assembly mounted to the housing and including a cartridge belt having a plurality of glove cartridges spaced along the cartridge belt, each glove cartridge having a glove releasably secured to the cartridge belt, wherein the glove cartridge assembly is operable to advance the cartridge belt and sequentially align each glove cartridge with the access aperture, and wherein negative pressure generated by the pressure system causes the glove of an aligned glove cartridge to inflate into the install cavity, and positive pressure generated by the pressure system seats the glove against a hand extended into the glove through the access aperture.

(85) B. A method of donning a glove on a hand, the method including activating a glove application system that includes a glove application module providing a housing that defines an install cavity, and a glove cartridge assembly mounted to the housing and including a cartridge belt having a plurality of glove cartridges spaced along the cartridge belt, each glove cartridge having a glove releasably secured to the cartridge belt. The method further including advancing the cartridge belt to align a glove cartridge with an access aperture defined in the housing and providing access into the install cavity, generating negative pressure in the install cavity with a pressure system and thereby inflating the glove into the install cavity, extending a hand into the glove inflated into the install cavity, generating positive pressure in the install cavity with the pressure system and thereby seating the glove against the hand, and releasing the glove from the cartridge belt.

(86) C. A glove application system that includes a mounting bracket that provides an access aperture, and a glove cartridge assembly mountable to the mounting bracket and including a cartridge belt having a plurality of glove cartridges spaced along the cartridge belt, each glove cartridge having a glove releasably secured to the cartridge belt with a fastening system, wherein

the glove cartridge assembly is operable to advance the cartridge belt and sequentially align each glove cartridge with the access aperture, and wherein extending a hand through the access aperture installs a glove from an aligned glove cartridge on the hand and simultaneously releases the glove from the cartridge belt.

(87) Each of embodiments A, B, and C may have one or more of the following additional elements in any combination: Element 1: wherein the pressure system comprises a device selected from the group consisting of a pressure piston, an air piston, a plunger, a wiper, a collapsible/deformable mechanism, a reversing compressor or pump, and any combination thereof. Element 2: further comprising a user interface activatable by a user to operate the glove application module. Element 3: wherein the housing comprises a cylindrical tube having a circular cross-section. Element 4: further comprising a mechanical actuator that is manually actuatable by a user to operate at least one of the pressure system and the glove cartridge assembly. Element 5: wherein the glove application module comprises a first glove application module and the glove cartridge assembly comprises a first glove cartridge assembly, the glove application system further comprising a second glove application module coupled to the first glove application module and providing a housing that defines an install cavity and an access aperture that provides access into the install cavity of the second glove application module, wherein the pressure system is operable to generate negative pressure and positive pressure in the install cavity of the second glove application module, and a second glove cartridge assembly mounted to the housing of the second glove application module and including a cartridge belt having a plurality of glove cartridges spaced along the cartridge belt, each glove cartridge of the second glove cartridge assembly having a glove individually packaged and releasably secured to the cartridge belt of the second glove cartridge assembly, wherein gloves included in the first glove cartridge assembly are of a first type and gloves included in the second glove cartridge assembly are of a second type different from the first type. Element 6: further comprising a user interface activatable by a user to select operation of the first or second glove application modules. Element 7: wherein the glove cartridge assembly comprises a supply spool and a recovery spool, and wherein the cartridge belt is extendable between the supply and recovery spools. Element 8: wherein the glove cartridge assembly is vacuum-packed and sealed within sterilized packaging prior to installation in the housing. Element 9: further comprising a sterilizing device positioned within the install cavity. Element 10: wherein each glove is releasably secured to the cartridge belt with a fastening system selected from the group consisting of a split O-ring, a releasable adhesive, a wax, a Velcro attachment, one or more mechanical staples, overlaying substrates, and any combination thereof.

(88) Element 11: wherein activating the glove application system comprises operating a user interface included in the glove application module. Element 12: wherein activating the glove application system comprises manually actuating a mechanical actuator included in the glove application system. Element 13: further comprising sterilizing an exterior of the glove with a sterilizing device positioned within the install cavity. Element 14: wherein the glove application system further includes a sterilization module coupled to the glove application module, the method further comprising extending the hand with the glove seated thereon into the sterilization module, and sterilizing the glove with a sterilizing device arranged within the sterilization module. Element 15: wherein releasing the glove from the cartridge belt comprises releasing the glove with the positive pressure.

(89) Element 16: wherein the fastening system comprises a split O-ring comprising a tubular structure that defines an interior, and an opening defined in the tubular structure to enable a wrist bead of each glove to be received within the interior to releasably secure the glove to the cartridge belt. Element 17: further comprising a mechanical actuator that is manually actuatable to operate the glove cartridge assembly and advance the cartridge belt. Element 18: further comprising an alignment guide through which the cartridge belt is extended to align the cartridge belt with the access aperture.

(90) Therefore, the disclosed systems and methods are well adapted to attain the ends and advantages mentioned as well as those that are inherent therein. The particular embodiments disclosed above are illustrative only, as the teachings of the present disclosure may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. Furthermore, no limitations are intended to the details of construction or design herein shown, other than as described in the claims below. It is therefore evident that the particular illustrative embodiments disclosed above may be altered, combined, or modified and all such variations are considered within the scope of the present disclosure. The systems and methods illustratively disclosed herein may suitably be practiced in the absence of any element that is not specifically disclosed herein and/or any optional element disclosed herein. While compositions and methods are described in terms of “comprising,” “containing,” or “including” various components or steps, the compositions and methods can also “consist essentially of” or “consist of” the various components and steps. All numbers and ranges disclosed above may vary by some amount. Whenever a numerical range with a lower limit and an upper limit is disclosed, any number and any included range falling within the range is specifically disclosed. In particular, every range of values (of the form, “from about a to about b,” or, equivalently, “from approximately a to b,” or, equivalently, “from approximately a-b”) disclosed herein is to be understood to set forth every number and range encompassed within the broader range of values. Also, the terms in the claims have their plain, ordinary meaning unless otherwise explicitly and clearly defined by the patentee. Moreover, the indefinite articles “a” or “an,” as used in the claims, are defined herein to mean one or more than one of the elements that it introduces. If there is any conflict in the usages of a word or term in this specification and one or more patent or other documents that may be incorporated herein by reference, the definitions that are consistent with this specification should be adopted.

(91) As used herein, the phrase “at least one of” preceding a series of items, with the terms “and” or “or” to separate any of the items, modifies the list as a whole, rather than each member of the list (i.e., each item). The phrase “at least one of” allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, the phrases “at least one of A, B, and C” or “at least one of A, B, or C” each refer to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

(92) The use of directional terms such as above, below, upper, lower, upward, downward, left, right, and the like are used in relation to the illustrative embodiments as they are depicted in the figures, the upward direction being toward the top of the corresponding figure and the downward direction being toward the bottom of the corresponding figure.

Claims

1. A glove application system, comprising: a mounting bracket mountable to a rigid structure and defining an access aperture in a plane perpendicular to a mounting surface of the mounting bracket, the mounting bracket extending laterally from the rigid structure to define an air gap interposing the mounting bracket and the rigid structure at the access aperture; and a glove cartridge assembly mountable to the mounting bracket and including a cartridge belt having a plurality of glove cartridges spaced along the cartridge belt, each glove cartridge having a glove releasably secured to the cartridge belt with a fastening system, wherein the glove cartridge assembly is operable to advance the cartridge belt and sequentially align each glove cartridge with the access aperture, and wherein extending a hand through the access aperture installs a glove from an aligned glove cartridge on the hand and simultaneously releases the glove from the cartridge belt.
2. The glove application system of claim 1, wherein the glove cartridge assembly comprises a pre-packaged and sterilized container including the plurality of glove cartridges.
3. The glove application system of claim 1, wherein the cartridge belt is wound on to a supply

spool mountable to the mounting bracket.

4. The glove application system of claim 3, further comprising a recovery spool mountable to the mounting bracket, wherein the cartridge belt is extendable between the supply and recovery spools.
 5. The glove application system of claim 1, wherein the fastening system is selected from the group consisting of a split O-ring, an adhesive, a wax, a hook and loop fastener attachment, one or more mechanical staples, overlaying substrates, and any combination thereof.
 6. The glove application system of claim 1, wherein the fastening system comprises a split O-ring comprising: a tubular structure that defines an interior; and an opening defined in the tubular structure to enable a wrist bead of each glove to be received within the interior to releasably secure the glove to the cartridge belt.
 7. The glove application system of claim 1, further comprising an alignment guide through which the cartridge belt is extended to align the cartridge belt with the access aperture.
 8. The glove application system of claim 1, wherein the glove cartridge assembly is motorized to advance the cartridge belt.
 9. The glove application system of claim 8, further comprising a user interface activatable by a user and configured for automated user operation of the glove cartridge assembly.
 10. The glove application system of claim 1, further comprising a mechanical actuator that is manually actuatable to operate the glove cartridge assembly and advance the cartridge belt.
 11. The glove application system of claim 10, wherein each manual actuation of the mechanical actuator advances and aligns a new glove cartridge of the plurality of glove cartridges with the access aperture.
 12. The glove application system of claim 1, wherein the cartridge belt comprises a single layer of material and the plurality of glove cartridges are coupled to the single layer of material, and wherein the glove in each glove cartridge is unsealed.
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